MOTURU MONISH

Software Engineer (SDE) • Full Stack & AI/ML Systems

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EDUCATION

Vignan's Institute of Information Technology • *Visakhapatnam, India* **Sept. 2023 – Present**

Bachelor of Technology in Computer Science (Data Science) • ***GPA: 8.64*** / ***10.0***

TECHNICAL SKILLS

- **Languages:** Python, Java, TypeScript, JavaScript, SQL, Kotlin
- **AI / ML & Edge:** TensorFlow Lite, ONNX Runtime, Scikit-learn, Gemma 4, LLM Integration, Pandas, NumPy, Prompt Eng.
- **Full Stack & Web:** React.js, React Native, Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, Tailwind CSS, MUI
- **Databases & Cloud:** PostgreSQL (Supabase), SQLite, MongoDB (familiar), Docker, Git, GitHub Actions, Playwright, Vercel

KEY ENGINEERING PROJECTS

Academic Monitor System | *React Native, Kotlin, TFLite, ONNX, BLE, Node.js, Supabase* **Jan. 2026 – Present**

- **Edge Biometric Pipeline:** Architected a 3-app offline-first attendance suite (Student, Faculty, Admin) using on-device MobileFaceNet TFLite (128-D embeddings, cosine threshold 0.60) paired with RSSI-gated P2P BLE handshakes (>-85 dBm) to eliminate proxy attendance without internet dependency.
- **Extreme Performance Optimization:** Engineered native Kotlin modules reducing face inference latency from ~6,000ms to **~200ms (25–40x speedup)**; deployed ArcFace ResNet-100 INT8 ONNX (512-D) on Admin client running at **52 FPS, 187ms latency, and 89MB RAM**.
- **Real-Time Sync & Automation:** Developed an 11-tab Admin console with Supabase WebSocket session locks preventing double-marking, and built an automated Playwright bot syncing records directly to legacy ERP portals.

Aequitas — AI Bias Mitigation Platform | *Python, Flask, Gemma 4, Perspective API, Supabase, Docker* **2026**

- **Fairness & Bias Auditing:** Built an AI auditing engine to detect and mitigate accent-based bias in automated video/audio interview scoring using Gemma 4 LLM and Perspective API across 200+ applicant profiles.
- **Post-Hoc Score Calibration:** Enforced a **<7% demographic parity variance** threshold via calibration algorithms without retraining; batch-ingested 500-record chunks into Supabase PostgreSQL and implemented SynthID-style cryptographic watermarking for audit report verification.

PhishGuard AI — Threat Detection Suite | *Python, FastAPI, Scikit-learn, React, MV3 Extension, SQLite* **Dec. 2025**

- **Hybrid ML Threat Scoring:** Deployed a Chrome Extension (MV3) backed by a Random Forest Classifier trained on 11,431 URLs (88 features, 200 estimators), computing risk scores via a 70% ML + 30% heuristic hybrid model.
- **Zero-Day Brand Protection:** Created `guard.js` DOM observer to detect form-action credential harvesting mismatches and Levenshtein distance checks (<2) guarding 13 global targets; provided explainable AI dashboard via FastAPI.

Skill Connect — Campus Collaboration Hub | *React, TypeScript, Supabase, Tailwind CSS, WebSockets* **Jan. 2025**

- **Full-Stack Platform:** Built an end-to-end platform (22 routes, 8 DB tables, ~4,150 lines of TypeScript) with real-time Supabase Postgres Changes WebSocket feeds, role-based matchmaking, and project lifecycle management.
- **Security & Trust Engine:** Enforced strict Row-Level Security (RLS) policies and database triggers for automated profiles; built a mandatory post-completion peer-review system recalculating member reputation scores.

LEADERSHIP & CAMPUS INVOLVEMENT

Web Development Lead | *E-Cell, Vignan's Institute of Information Technology* **Aug. 2024 – Dec. 2025**

- Directed web infrastructure development for official entrepreneurship portals, ensuring 99.9% uptime and handling registrations for **500+ participants**; mentored a team of 5 junior developers in modern React and Git workflows.

Campus Ambassador | *GUESSS India • IIT Mandi* **Aug. 2025 – Dec. 2025**

- Served as primary regional liaison collecting 100+ validated survey responses to evaluate collegiate entrepreneurial intentions across South India.

HONORS & ACHIEVEMENTS

- **National Finalist (Top 0.6% of 1,000+ Teams):** National Entrepreneurship Challenge (NEC) 2025, **IIT Bombay**.